The Homunculus Resurrected: Functional Semi-Permeability and the Feeling of Unified Agency

Forward

The philosophical idea that the mind is separate from its physical substrate has often been attacked as naive dualism. It is caricatured as a tiny person living inside the brain, a homunculus. Yet the feeling is almost universal that I am separate from my body, that this is what it feels like to be conscious. The religious experience is of having a soul. I cannot help but think that at least in part the historical scientific conflict with religion explains the dismissive caricature. Ironically, as we have gained scientific knowledge, the mind outside the body is starting to look more reasonable. We now know that simple organisms outsource memory to the environment in the form of chemical trails. As humans we do it all the time in the form of paper notes, books, and the like. When making a calculation we do not hold everything in memory. We calculate through an external medium. Knowledge becomes a collective process as we share our records. The process is almost universal and is called swarm intelligence. We are born with the ability to acquire knowledge, but it is acquired from the environment through physical interaction with the environment and from people and records. Central to that process is the homunculus, the feeling of being a separate thing.

There is a common experience of the process when you raise puppies. Through repetition puppies learn they have a name. When you call a puppy they do not all respond but do so according to their assigned name. The same puppy that was chasing its tail a minute before, as if the tail was a separate thing from them, stops and runs toward you. The tail does not come to you. The mind of the puppy does. The thing that has an identity separate from everything in the environment, including the tail. When you train an animal you are not training its body. You are training the identity. That is why the concept of the “horse whisperer” is so effective. Negotiation and a relationship between your identity and the horse’s identity turns out to be more effective than force or conditioned reflex. This distinction is crucial to what scientists call behavioral flexibility. Animals are not wet robots. They have the evolved ability to adapt to novel conditions, something that is proving very difficult to reproduce in robotics. The rest of this document explores the little person inside us all.

I am not defending the philosophical homunculus fallacy, a literal little person inside the brain. I am defending the common human intuition that there is a unified “someone” to whom experience happens. My argument is that this intuition is not a scientific mistake but an evolved feature of cognition.

This essay offers an architectural account of why a vivid, concentrated feeling of unified self and agency appears at certain layers of organized systems. It does not address the deeper question of why subjective experience exists at all. That question lies beyond current scientific reach. The more limited claim pursued here is that the homunculus feeling is a predictable consequence of functional partial isolation at executive layers, and that this isolation serves coordination across scales even while remaining dependent on the layers beneath it.

Grounding the argument

Consciousness is not located in any single function or layer of organization. It appears in degrees across the system as a whole. The vivid, concentrated sense of a unified self and of being the locus of decision and agency arises most strongly at the executive level. This concentration occurs because the executive layer maintains a degree of functional partial isolation from the more immediate activity of subordinate processes. That isolation is not produced by a physical membrane or enclosure but by the functional demands of cross-scale coordination and priority-setting. The resulting phenomenology, the felt presence of a central controller, is therefore an emergent feature of the architecture rather than evidence of an actual central entity.

The pattern of regulated exchange without total separation appears already at the level of the cell. The membrane creates internal coherence while remaining open to material and energetic flows from outside. This same logic of functional semi-permeability scales through successive layers of organization. Lower layers remain in relatively direct contact with physical and energetic conditions. Higher layers operate with greater abstraction and temporal reach. The partial isolation that develops at these higher levels enables integration across multiple subordinate processes, yet it also generates the distinctive first-person character of executive experience. What feels like a homunculus is the lived quality of operating from within a layer whose functional role requires a workable degree of separation from the full detail of lower-level activity.

Because consciousness is graded rather than localized, lower layers retain their own forms of responsiveness and coupling. Executive modulation does not replace these lower forms of organization but selects among and constrains them. The strong sense of unified agency that accompanies this modulation depends on the continued functioning of the layers beneath it. When those layers lose capacity, drift in their priorities, or withdraw cooperation, the higher layer’s effective influence declines even when its formal authority remains intact. Top-down direction is therefore real in its consequences but always conditional on the stability and alignment of the underlying architecture. It does not require a central controller to be effective, yet neither does it dissolve into purely bottom-up emergence.

The deeper and more persistent error lies not in the homunculus feeling itself but in the inference drawn from that feeling. The assumption that coherent agency ultimately requires a genuine center of control has survived the rejection of the classical homunculus picture. Once the system is described as layered swarms connected by functional semi-permeability, the appearance of centralization can be recognized as an interface phenomenon rather than as evidence of an underlying centralized architecture. The executive layer produces a compressed representation of unified control because that compression serves coordination across scales and across time. The compression is useful precisely because it is lossy. Treating it as a literal description of the system’s organization converts a functional interface into an ontological claim.

This reframing also alters how we understand the persistence of the central-controller intuition. The architecture reliably generates a concentrated sense of agency at certain layers. It is therefore unsurprising that the intuition reappears across widely different frameworks. The feeling is not an accidental cultural overlay but a structural consequence of how partial isolation at higher layers makes certain forms of coordination and self-description possible.

Randomness and variability continue to play a generative role within this architecture. They exert ongoing pressure against the higher layers’ tendency to stabilize into excessive interiority. When intermediate filtering mechanisms further insulate executive processes from lower-level feedback, the system risks becoming overly self-referential. Thermodynamic accounts of information processing show that logically irreversible operations, including the effective erasure or heavy compression of lower-level detail, carry a minimum physical energy cost. This cost provides a physical reason why higher layers cannot arbitrarily sever contact with generative variability originating at more physical scales without consequence. Randomness is not merely statistical fluctuation or external perturbation. At the substrate level it supplies variability that higher layers cannot fully predict or internalize. In this respect it functions as a continuing corrective that keeps functional semi-permeability from hardening into closure.

The same layered architecture that produces the concentrated sense of self at the executive level therefore remains open to disruption from below and from irreducible physical variability. Top-down influence remains possible and often effective, yet it never becomes self-sufficient. The system can sustain coordinated behavior across scales without requiring a center, while still allowing higher layers to exert real directive effects when conditions permit. The homunculus feeling is the subjective correlate of inhabiting one of those higher layers. The error lies in mistaking that correlate for evidence that the underlying organization must itself be centralized.

This account departs from both classical pictures that posit a central controller and from stronger enactivist or autopoietic views that treat higher-level directive influence as largely epiphenomenal. It preserves a functional role for top-down coordination while insisting that such coordination remains dependent on the continued operation and alignment of the layers beneath it. The partial isolation that makes executive function possible is itself a product of the system's architecture rather than a metaphysical remainder. In this sense the resurrected homunculus is not a return to an earlier error but a recognition of what certain layers of organized, semi-permeable systems naturally produce when they perform coordinating work.

Distributed Tools and Functional Semi-Permeability: Bees, Crows, and Neural Swarms

The central architectural feature that allows complex coordination without a central controller is functional semi-permeability between distinct regions or components. This pattern appears at multiple scales. At the cellular level, membranes regulate exchange while preserving internal coherence. At the neural level, distinct brain regions maintain degrees of functional separation while still exchanging information. This separation allows local populations of neurons to self-organize into effective swarms while also participating in larger-scale coordination. The same logic scales outward to interactions between individuals in a colony or between an organism and its environment.

In honey bee colonies, individual bees interact according to relatively simple rules shaped by local conditions and signals from other bees. No single bee maintains a model of the colony's overall state or issues top-down commands. Yet the colony as a whole regulates foraging, maintains temperature, and adjusts division of labor in response to changing conditions. The functional boundaries between individual bees and between task groups allow local self-organization while still permitting colony-level coordination. The system remains open to environmental feedback at every point.

When we examine individual animals capable of tool use, such as members of the genus *Corvus*, the same principle of functional semi-permeability appears at the neural level. Different brain regions can operate with varying degrees of independence while still communicating. This allows local neural populations to self-organize around specific aspects of a problem—spatial relationships, object properties, motor sequences—without every element having to be held in a single central workspace. Studies of corvid cognition show that these birds can solve novel multi-step problems and manufacture tools, yet they do so with limited working memory capacity and without the extensive ability to offload intermediate results into the environment that humans use. The functional separation between neural regions permits effective problem-solving within these constraints, but it also imposes clear limits on how far and how stably a decoupled goal can be maintained.

In humans, and particularly in domains requiring sustained abstract work such as mathematics, studies show that cognitive tools and representations are distributed across multiple brain regions rather than concentrated in a single executive workspace. Different regions appear to handle distinct aspects of a problem—spatial visualization, symbolic manipulation, memory retrieval, motor planning for writing or drawing—and these regions interact through regulated channels rather than through total integration. This functional semi-permeability allows local neural swarms to self-organize around their specialized functions while still contributing to larger-scale problem-solving. Experts further extend this architecture by using external representations—diagrams, notation, written steps—that function as temporary extensions of the system. These external elements reduce the load on any single internal region and allow the overall process to maintain coherence across longer timescales and greater complexity.

The comparison across these cases suggests that the underlying process is continuous rather than fundamentally different in kind. In each instance, functional semi-permeability between components—whether individual bees, neural regions, or brain-plus-environment systems—enables local self-organization while still supporting coordination at larger scales. The bee colony achieves collective regulation through interactions among many simpler units. The crow achieves individual tool use through interactions among neural regions that retain a degree of independence. Human experts achieve more extended problem-solving by adding environmental offloading to the same basic pattern of distributed, semi-permeable resources.

What changes across these systems is primarily the degree of offloading and the stability with which decoupled goals can be maintained. When environmental memory storage is limited, as it is for crows, the higher-level coordination remains more tightly constrained by immediate perceptual and motor feedback. When external representations and distributed internal storage are available, as they are for human experts, the same architectural logic can support more complex and sustained decoupling without requiring any single region or individual to hold the entire problem state centrally.

This continuity matters for how we interpret the persistent intuition of a central controller. The feeling of unified agency arises most strongly when a higher layer—whether a neural region or a cultural practice—achieves enough functional isolation to coordinate across multiple subordinate processes. That isolation is real and useful, but it is always partial and conditional. It depends on the continued operation of the semi-permeable boundaries that allow both separation and exchange. When those boundaries are disrupted or when the lower-level components withdraw effective support, the appearance of centralized control weakens even if formal structures remain in place.

The same architectural feature that permits self-organization of neural swarms within swarms also limits how far any higher layer can drift into excessive interiority. Because the boundaries remain functionally permeable, variability generated at more physical scales continues to exert pressure on higher-level models. This pressure does not eliminate the utility of functional isolation, but it prevents that isolation from becoming total. The result is a system that can produce coordinated, goal-directed behavior across scales without ever requiring a single point of comprehensive control.

The human advantage, exploiting the margins

Corvid cognition was mentioned early in reference to tool building, but an interesting finding is that they can solve some problems more efficiently than children. In a way this somewhat surprising finding illustrates lossy compression. A crow cannot afford years of inefficient, exploratory tool use. It needs the evolved tool holder to work reliably from early on because the environment will not subsidize a long developmental runway. Human children can be wildly inefficient, wrong, and playful for years precisely because caregivers, social structures, and cultural institutions buffer the survival cost of those errors. The openness is not free. It is paid for by the social environment. Paying for the compression up front has its cost in long-term adaptability. Culture is not just what gets stored in the tool holder across generations. It is the error-tolerance infrastructure that makes the open pipeline possible in the first place. Remove the social buffering and you would expect selection pressure to close the pipeline down, favoring faster, narrower, more algorithmic development.

Environments with very low error tolerance, whether due to resource scarcity, instability, or harsh social conditions, push development toward faster closure of that open pipeline. Kids in high-stress environments often show earlier, more rigid cognitive consolidation. That is not a deficit in any simple sense. It is the system responding rationally to reduced error tolerance cost. Crows do cache and learn from each other to some degree. The difference is in how plastic and socially extended the revision process is. That is not to say that a larger, more powerful brain is not key. It addresses what makes exploration possible and worth the cost. This relates indirectly to the concept of genius. High intelligence is a necessary but insufficient condition for genius. Genius is high intelligence plus the generation of large numbers of possible solutions and the ability to sort through them rapidly for increased coherence to observation. There are many more highly intelligent people than geniuses. Genius trades efficiency for imagination.

Whether consciousness ultimately proves to be an emergent property, a fundamental feature of nature, or something we have not yet imagined, the evidence explored throughout this series suggests that the experience of being a distinct self is not an obstacle to a scientific understanding of intelligence. It may instead be one of evolution's most important adaptations.

This essay forms part of a series examining variability as the default state of systems and the minimum-energy, lossy constraints that produce temporary coherence and functional capacity across scales. Related discussions appear in the notes on randomness as structured potential, coherent signaling, probabilistic agency, mythos cognition, correspondence costs, and the layered architecture of agency. The generation of variants is so central in this series of essays that the reader may think that is the thesis itself. It is important to remember that system stability must be historically robust: strong enough to absorb or generate variation without erasing or scrambling the accumulated, path-dependent structure that constitutes the system's identity and capabilities. As with life in general, reproductive fidelity is central to preserving anything for variants to emerge from and be selected from. Readers may enter at any point.